2023 Certificates: Install How-To

Introduction

Later this month (June 2026) the Security Certificates issued by Microsoft in 2011 will expire. These certificates are stored in UEFI firmware on all PCs that had once ran Windows 8 or newer.

In 2023 Microsoft released an updated file (shimx64.efi) that is used to perform the first part of Secure Boot verification on Linux PCs with Secure Boot in an enabled state.

Almost all of the existing Operating Systems that used these 2011 Certificate will still boot.[1]

The reason is the subtlety of the differences in definitions of ‘Expired’ versus ‘Revoked’ words. Microsoft has stated that at some undefined point in the future they will Revoke the 2011 certs.

This expiration does NOT have any effect on Linux drivers or libraries signed by your PC (by Debian’s public certificate) or internal or external Hardware Manufacturer’s existing Security Certificates. Their certificates do NOT depend on the Microsoft certificates to achieve CA Root authenticity in any way.

Installing the new certificates in Linux

1. Install the Debian package fwupd (2.0.10 or newer) by typing in terminal:

   `sudo apt install fwupd`

2. Update the local database of firmware updates by typing in terminal:

   `sudo fwupdmgr get-updates`

3. Run the ‘update’ option in terminal:

   `sudo fwupdmgr update`

4. You will be prompted two times to Perform operation? [Y|n]:
   Example is below.

```
Upgrade UEFI CA from 2011 to 2023?

This updates the 3rd Party UEFI Signature Database (the "db") to the latest
release from Microsoft.It also adds the latest OptionROM UEFI Signature
Database update.

UEFI CA and all connected devices may not be usable while updating.

Perform operation? [Y|n]: Y
Waiting…                 [***************************************]
Successfully installed firmware

Upgrade UEFI dbx from 20220801 to 20250902?

This updates the list of forbidden signatures (the "dbx") to the latest
release from Microsoft.

Some insecure versions of the IGEL bootloader were added, due to a security
vulnerability that allowed an attacker to bypass UEFI Secure Boot.

Perform operation? [Y|n]:
```

Press [enter] to answer Y to each.

1 **When Secure Boot certificates expire on Windows devices** - https://support.microsoft.com/en-us/topic/when-secure-boot-certificates-expire-on-windows-devices-c83b6afd-a2b6-43c6-938e-57046c80c1c2

Updating shimx64.efi

The 1.49 version of `shim-signed.efi` is in Debian Sid. In MX Linux 25 version 1.47 is current. Grub is signed by Debian so no update of it is needed. To update it type in terminal:

```
sudo apt install shim-signed
```

Verify certificates installation

To verify the Certificates type in terminal: `mokutil -db -short`. Example below.

```
Terminal - mike@MX2
File  Edit  View  Terminal  Tabs  Help
mike@MX25Xfce:~
$ mokutil --db --short
15254b199b Dell Inc. UEFI DB
46def63b5c Microsoft Corporation UEFI CA 2011
580a6f4cc4 Microsoft Windows Production PCA 2011
b5eeb4a670 Microsoft UEFI CA 2023
3fb39e2b8b Microsoft Option ROM UEFI CA 2023
```

Notice 2011 and 2023 certificates are present. This is normal.

To verify the shimx64.efi signing

In terminal enter:

```
sbverify --list /boot/efi/EFI/MX/shimx64.efi
```

Below is typical output before shimx64.efi is signed with the 2023 certificates:

```
mike@MX25Xfce:/boot/efi/EFI/MX
$ sbverify --list shimx64.efi
warning: data remaining[831016 vs 957136]: gaps between PE/COFF sections?
signature 1
image signature issuers:
 - /C=US/ST=Washington/L=Redmond/O=Microsoft Corporation/CN=Microsoft Corporation UEFI CA 2011
image signature certificates:
 - subject: /C=US/ST=Washington/L=Redmond/O=Microsoft Corporation/CN=Microsoft Windows UEFI Driver Publisher
   issuer:  /C=US/ST=Washington/L=Redmond/O=Microsoft Corporation/CN=Microsoft Corporation UEFI CA 2011
 - subject: /C=US/ST=Washington/L=Redmond/O=Microsoft Corporation/CN=Microsoft Corporation UEFI CA 2011
   issuer:  /C=US/ST=Washington/L=Redmond/O=Microsoft Corporation/CN=Microsoft Corporation Third Party Marketplace Root
```

Source - https://learn.microsoft.com/en-us/windows-hardware/manufacture/desktop/windows-secure-boot-key-creation-and-management-guidance?view=windows-11#14-signature-databases-db-and-dbx

“For devices that are intended to support Linux, other 3rd party UEFI apps, or include special hardware requiring Option ROMs, OEMs may ship devices with this alternative configuration:”

After update this is what the Cerificate Store in the TPM should look like:

PK	OEM or Microsoft PK
KEK	Microsoft Corporation KEK 2K CA 2023
DB	Windows UEFI CA 2023 Microsoft UEFI CA 2023 Microsoft Corporation UEFI CA 2011 Microsoft Option ROM UEFI CA 2023
DBX	Latest dbx package

Windows 11

The concept of the phrase ‘the clock is ticking’ is in play. Procrastinating past the time Microsoft does the revocations WILL have consequences. A quote from Microsoft:

“OEMs must ship the device with the Windows UEFI CA 2023-signed Windows Boot Manager, both on the device and in bootable media provided with the device. This ensures that the device uses the Windows Boot Manager that will continue to be serviced after the 2011 certificates expire.”

Links

Secure Boot Certificate Changes in 2026 Red Hat Customer Portal

Windows Secure Boot certificate expiration and CA updates Microsoft Support

SecureBootCAChanges Debian WiKi

When Secure Boot certificates expire on Windows devices